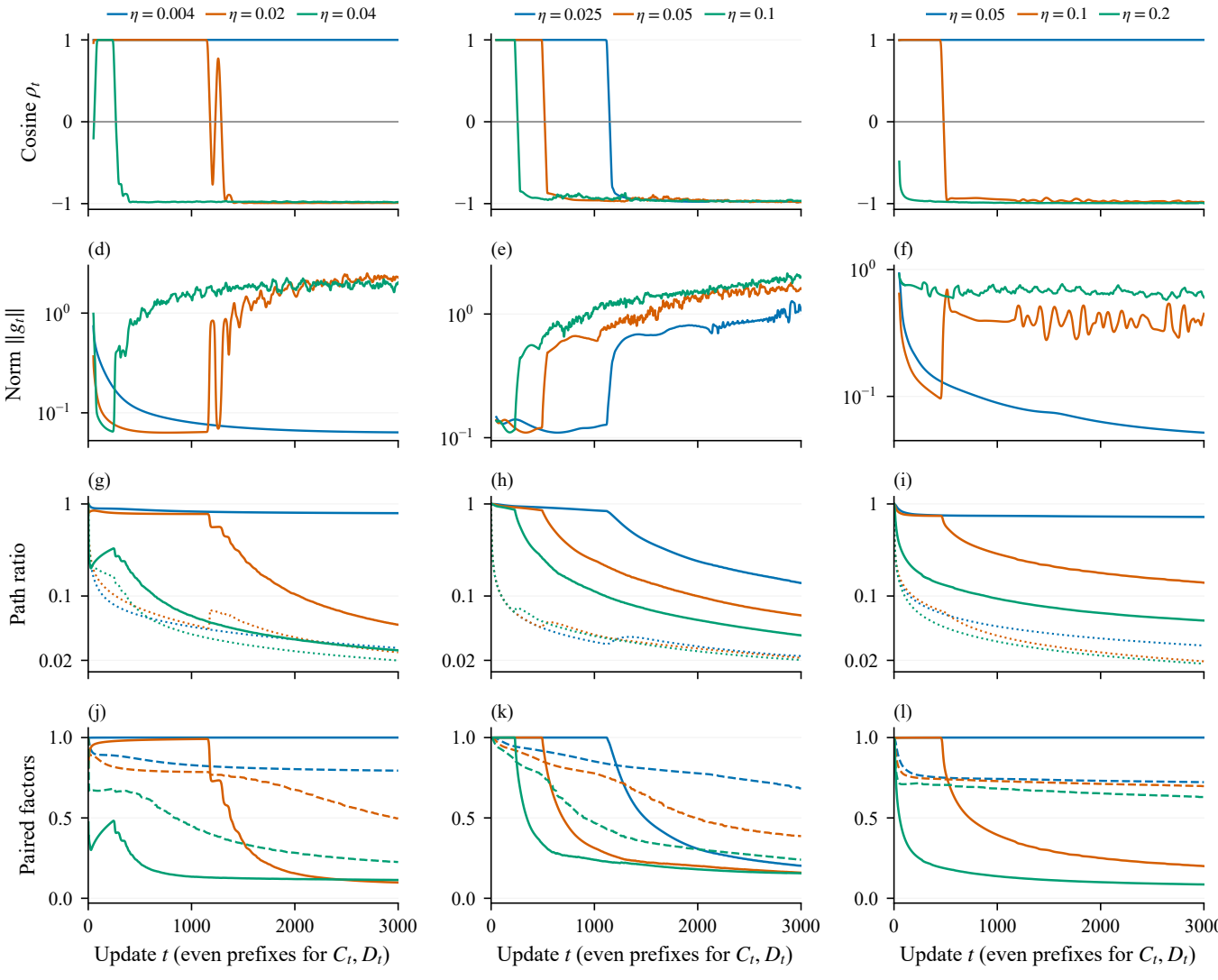

(a) CIFAR-10 / MLP
(b) CIFAR-10 / ResNet-32
(c) WikiText-2 / Transformer
η = 0.004
η = 0.02
η = 0.04
η = 0.025
η = 0.05
η = 0.1
η = 0.05
η = 0.1
η = 0.2
Cosine ρ_t
Norm ‖g_t‖
Path ratio
Paired factors
(d)
(e)
(f)
(g)
(h)
(i)
(j)
(k)
(l)
Update t (even prefixes for C_t, D_t)
R_t
R_t^⊥
C_t: within-pair retention
D_t: between-pair retention